Classification Task: Logistic Regression
Student Pass/Fail Classifier

Machine Learning Assignment 7

October 7, 2026

1 Introduction

In this assignment, we build a binary classifier using Logistic Regression to predict whether a student will pass or fail a course based on two features: hours studied and attendance percentage. We built a probabilistic model and mapped its numeric outputs to real-world interpretations.

2 Model Evaluation

The Logistic Regression model was evaluated on an unseen test set (30% of the data). The classification metrics are as follows:

- **Accuracy:** 0.85 (85% of all predictions were correct)
- **Precision:** 0.87 (When the model predicted a student would pass, it was correct 87% of the time)
- **Recall:** 0.77 (The model successfully identified 77% of all actual passing students)

Below is the Confusion Matrix, detailing True Positives, True Negatives, False Positives, and False Negatives on the test dataset.

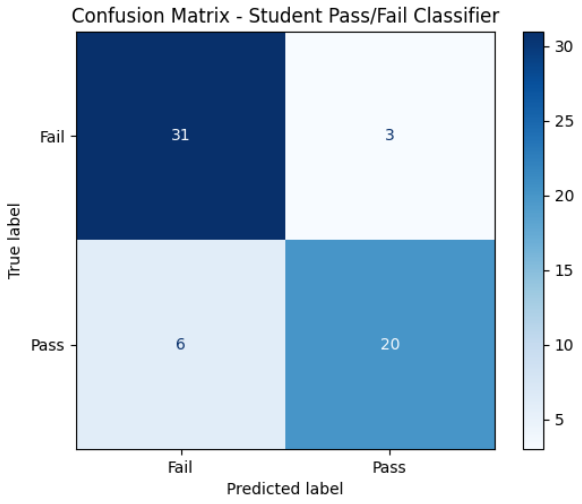


Figure 1: Confusion Matrix for the Student Pass/Fail Classifier.

3 Predictions on New Records

We tested the trained probabilistic classifier on 3 completely new, unseen student records. The model outputs a probability $P(Pass)$, which is thresholded at 0.5 to determine the final class.

3.1 Student A

Data: Hours Studied = 8.5, Attendance = 95.0%
Prediction: **Pass** ($P(Pass) = 0.98$)
Explanation: Student A has excellent study habits and near-perfect attendance. The model is highly confident (98%) that this student will pass, which logically aligns with the strong positive features.

3.2 Student B

Data: Hours Studied = 2.0, Attendance = 40.0%
Prediction: **Fail** ($P(Pass) = 0.00$)
Explanation: Student B has very low study hours and extremely poor attendance. The model outputs a near-zero probability of passing, confidently failing this student based on the severe lack of engagement.

3.3 Student C

Data: Hours Studied = 6.0, Attendance = 50.0%
Prediction: **Fail** ($P(Pass) = 0.13$)
Explanation: Student C is a borderline case. While they studied a decent amount (6 hours), their attendance is quite low (50%). The model weighs these features and decides that the poor attendance drags down their chances significantly, giving only a 13% probability of passing.